Jake HONG Zehao

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Category: Fresh Graduate (Master's) | **Exp:** 2 Years | **Visa:** IANG

Language: English (Fluent), Mandarin (Native), Cantonese (Proficient L/R/W; Active Speaking Learner)

Passionate **Master's Candidate in AI and Business Analytics** with a specialty in **Application Development**, **Data Analytics** and **Artificial Intelligence**.
Proficient in **Java, Python and Multi-database systems (SQL/NoSQL)**.
Dedicated to enhancing **User Experience** through data-driven automation and scalable system design. Eager to contribute to digital innovation ecosystem.

EDUCATION BACKGROUND

Lingnan University *Hong Kong*
Master of Science in AI and Business Analytics *September 2025 – July 2026 (Expected)*

- **Core Curriculum**: Blockchain, Big Data Analytics, Machine Learning, Deep Learning, Digital Infrastructure, Business Intelligence.
- Exploring and transforming data into actionable insights.

TECHNICAL SKILLS

- **Software Development**: Java (Spring Boot), Python (FastAPI/Django), JavaScript (ES6+), HTML5/CSS3.
- **Data & Cloud**: SQL (**MySQL, Oracle, MS SQL Server**), AWS (EC2/S3), Docker, Kubernetes, Apache Spark.
- **Business Skills**: Problem Framing, KPI Design, A/B Testing, Storytelling, Reporting
- **Techniques**: NLP, Machine Learning, Data Modeling, Data Visualization, Disaster Recovery.
- **Tools**: pandas, NumPy, scikit-learn, Power BI, GitHub, Excel

PROFESSIONAL EXPERIENCE

Beijing Gamutsoft IT Service Inc. *Monitoring Support Engineer* *May 2024 – February 2025*

- **System Reliability**: Managed 24/7 high-availability surveillance for enterprise-grade digital infrastructure, ensuring 99.9% uptime through real-time predictive analytics.
- **High-Pressure Problem Solving**: Resolved critical system anomalies using standardized incident workflows, effectively curbing potential financial and operational risks.
- **Cross-Border Collaboration**: Liaised with global telecom providers to optimize network latency across international regions, supporting seamless digital service delivery.

Shenzhen Chaolunjia Technology Co., Ltd. *Systems Engineer* *March 2023 – March 2024*

- **Automation & Efficiency**: Designed Python-based automation frameworks for system provisioning, reducing deployment time by 30% and enhancing operational agility.
- **Infrastructure Security**: Performed deep root-cause analysis on network vulnerabilities and implemented security patches to safeguard corporate data integrity.
- **Agile Integration**: Orchestrated the end-to-end integration of new hardware and software systems, ensuring compatibility with existing digital service architectures.

RELEVANT INTERNSHIP EXPERIENCE

Guangdong Teddy Intelligent Tech Co., Ltd *Data Analyst Intern* *November 2020 – February 2021*

- **User-Centric Analytics**: Analyzed 50k+ campus transaction records to identify consumption trends; created interactive Tableau dashboards to drive operational decision-making.
- **Product Optimization**: Conducted A/B testing feedback loops through user surveys to refine product interface features and peak-hour service allocation.

China United Communications (Unicom) *Software Engineer Intern* *October 2019 – June 2020*

- **Internal System Dev**: Built a Java-based points management platform, optimizing the backend logic for faster query response and improved staff workflow transparency.
- **Database Tuning**: Structured relational schemas in MS SQL Server and optimized query performance to analyze ARPU metrics and customer usage patterns.

KEY PROJECT EXPERIENCES

Titanic Survival Prediction ***Machine Learning****: Python, scikit-learn, Pandas, NumPy*

- Developed a machine learning model to predict passenger survival on the Titanic, performing data cleaning, EDA, feature engineering, and evaluating multiple models, achieving strong performance with key predictors identified through a full end-to-end pipeline.

IMDb Sentiment Classification ***AI / NLP****: Transformers, BERT, RoBERTa, Llama 3.2*

- Built a dual-model NLP system to classify movie review sentiment using ML and LLMs (SVM, BERT, RoBERTa-Large, and Llama 3.2 with LoRA fine-tuning), where RoBERTa achieved the best accuracy and the LoRA model demonstrated high efficiency, with full responsibility for model implementation, hyperparameter tuning, and evaluation.

Automatic Diagnosis Through Electrocardiograms (ECG) ***Signal Processing & ML****: Python, SVM, CNN*

- Enhanced heart-disease diagnosis accuracy through ECG signal analysis by preprocessing signals (removing baseline wander), comparing 28 R-peak detection algorithms, evaluating heartbeat extraction performance, and exploring SVM and CNN models for classification, ultimately building a reproducible signal-processing and diagnostic pipeline with high detection accuracy and robust performance.

Employment Recommendation System ***Full-Stack****: Django, MySQL, Python*

- Developed a scalable match-making platform using collaborative filtering algorithms; successfully handled a database of 13,000+ job records with optimized SQL indexing.

Campus Consumption Behavior Analysis ***Data Analytics****: Python, Tableau / Power BI, SQL*

- Analyzed student consumption patterns on campus by preprocessing transaction data, performing visual analytics on canteen usage distribution and weekday vs. weekend dining habits, and integrating structured survey feedback to combine quantitative and qualitative insights, ultimately delivering actionable recommendations for canteen optimization and resource allocation.

Shell One-Click Deployment Framework ***DevOps****: Linux Automation, Shell Scripting*

- Implemented "Infrastructure as Code" concepts by creating modular scripts for rapid environment replication and automated failover in distributed systems.

Credit Card Fraud Detection System ***Fintech Security****: Scikit-learn, Python*

- Engineered a risk monitoring model using Decision Tree classifiers to detect fraudulent transactions in imbalanced datasets, emphasizing precision and security.

Walmart Customer Segmentation ***Data Science****: KMeans, Marketing Analytics*

- Applied the Elbow Method and KMeans clustering to segment customers based on purchasing behavior, providing data-driven insights for targeted digital marketing.

CERTIFICATES & AWARDS

- **Certifications:** Python Technology Application Engineer (MIIT), Google Data Analytics Professional.
- **Selected Awards:** 2nd Prize in Provincial English Writing Competition.